THE GALACTIC CANVAS

3D Website AI Generation — JSON Design Prompt

Use this prompt with an AI coding/website generator to redesign the existing Three.js website as a cinematic, scroll-driven space journey.


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{
  "project": {
    "name": "The Galactic Canvas",
    "type": "Interactive 3D Educational Website",
    "theme": "Cinematic Space Journey",
    "technology": "HTML, CSS, JavaScript, Three.js",
    "primary_interaction": "Scroll-driven 3D storytelling",
    "goal": "Transform the existing 3D Milky Way and Solar System website into a cinematic, interactive journey from the Milky Way to the
  },
  "important_instruction": {
    "existing_project": "Use the uploaded The Galactic Canvas HTML as the starting point.",
    "preserve": [
      "Three.js based 3D experience",
      "existing astronomical data",
      "existing planet information",
      "existing interactive Learn More functionality",
      "dark space visual language",
      "responsive behavior",
      "reduced-motion accessibility option"
    ],
    "do_not": [
      "Do not completely rebuild the project from scratch unless necessary.",
      "Do not remove the existing astronomical information.",
      "Do not show the entire Solar System extremely large on the Milky Way introduction page.",
      "Do not use generic flat website sections instead of the 3D experience.",
      "Do not make all planets use the same camera position.",
      "Do not overload the screen with text."
    ]
  },
  "overall_experience": {
    "concept": "The user should feel like they are physically travelling through space.",
    "journey": [
      "Milky Way Introduction",
      "Milky Way Information",
      "Space Jump / Nebula Transition",
      "Solar System",
      "Sun",
      "Mercury",
      "Venus",
      "Earth",
      "Mars",
      "Jupiter",
      "Saturn",
      "Uranus",
      "Neptune"
    ],
    "visual_style": [
      "cinematic",
      "minimal",
      "premium",
      "scientific",
      "immersive",
      "dark",
      "futuristic",
      "editorial"
    ]
  },
  "design_system": {
    "background": "#03030A",
    "primary_text": "#EEF1FF",
    "secondary_text": "#A9B0D0",
    "accent": "#8EA2FF",
    "panel_background": "rgba(14,16,34,0.62)",
    "border": "rgba(160,175,255,0.22)",
    "heading_font": "Space Grotesk",
    "body_font": "Inter",
    "ui_style": "minimal glassmorphism",
    "border_radius": "14-18px",
    "visual_priority": "3D object first, information second"
  },
  "page_01": {
    "name": "Milky Way Introduction",
    "purpose": "Create a strong cinematic opening.",
    "visual": "Show the Milky Way galaxy as the main background environment.",
    "solar_system_visibility": "Do not show the Solar System prominently on this page.",
    "hero_text": {
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    "title": "THE GALACTIC CANVAS",
    "title_style": "very large, bold, centered",
    "subtitle": "A visual journey through our galaxy, solar system and planets.",
    "scroll_hint": "SCROLL TO EXPLORE"
  },
  "layout": "center aligned",
  "animation": {
    "title": "fade in + slight scale up",
    "subtitle": "fade in after title",
    "background": "slow galaxy particle movement",
    "scroll_hint": "subtle pulsing animation"
  }
},
"page_02": {
  "name": "The Milky Way",
  "purpose": "Explain the galaxy before entering the Solar System.",
  "camera": "slow cinematic movement through the Milky Way",
  "headline": "THE MILKY WAY",
  "description": "A vast barred spiral galaxy containing billions of stars, including our Sun.",
  "information_sequence": [
    {"label": "GALAXY TYPE", "value": "BARRED SPIRAL"},
    {"label": "LOCATION", "value": "ORION ARM"},
    {"label": "SCALE", "value": "Approximately 100,000 light-years across"},
    {"label": "OUR HOME", "value": "The Solar System"}
  ],
  "text_animation": "Each information block fades in, stays visible briefly, then transitions to the next.",
  "rule": "Do not display every information block at once."
},
"page_03": {
  "name": "Space Jump Transition",
  "purpose": "Create a cinematic transition from the Milky Way to the Solar System.",
  "transition_style": "space jump through a nebula",
  "sequence": [
    "Camera identifies the Solar System location inside the Milky Way.",
    "Galaxy particles gradually accelerate.",
    "Stars begin stretching toward the viewer.",
    "Nebula particles appear around the camera.",
    "Camera rapidly travels forward through space.",
    "Create a subtle hyperspace / space-jump effect.",
    "Slow the camera down.",
    "Reveal the Solar System in the distance."
  ],
  "effects": [
    "particle acceleration",
    "star streaks",
    "nebula clouds",
    "depth illusion",
    "subtle motion blur",
    "camera zoom",
    "smooth easing"
  ],
  "important": "The transition should feel like physical travel through space, not a simple fade."
},
"page_04": {
  "name": "Solar System",
  "purpose": "Introduce the Solar System as its own dedicated experience.",
  "camera": {
    "starting_shot": "medium-wide cinematic shot",
    "scale": "Solar System should be clearly visible but not oversized",
    "composition": "balanced negative space around the system"
  },
  "headline": "THE SOLAR SYSTEM",
  "subheading": "One star. Eight planets. Countless smaller worlds.",
  "information": [
    "ONE STAR",
    "EIGHT PLANETS",
    "BILLIONS OF SMALLER OBJECTS",
    "4.6 BILLION YEARS OLD"
  ],
  "interaction": [
    "Allow the user to rotate the Solar System.",
    "Allow hovering over planets.",
    "Highlight the selected planet.",
    "Allow clicking a planet to travel toward it.",
    "Allow scroll to move the camera toward the Sun."
  ],
  "important": "Do not keep the Solar System as a tiny background object after this page. It becomes the main subject."
},
"page_05": {
  "name": "Sun",
  "purpose": "Give the Sun a dedicated cinematic page.",
  "transition": "Camera travels from the Solar System toward the Sun.",
  "camera": {
    "shot": "close-up / half-shot",
    "sun_screen_coverage": "50-70 percent",
    "composition": "Sun on one side with information on the other"
  },
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  "headline": "THE SUN",
  "subheading": "The star at the heart of our Solar System.",
  "data": [
    {"label": "DIAMETER", "value": "1,392,700 km"},
    {"label": "AGE", "value": "4.6 billion years"},
    {"label": "SURFACE TEMPERATURE", "value": "≈5,500 °C"},
    {"label": "CORE TEMPERATURE", "value": "≈15 million °C"}
  ],
  "interaction": {
    "type": "interactive hotspots",
    "hotspots": ["Sunspots", "Solar Flares", "Solar Wind", "Corona"]
  },
  "animation": [
    "solar surface movement",
    "subtle glow",
    "slow rotation",
    "hotspot pulse"
  ]
},
"planet_experience": {
  "general_rule": "Every planet must have its own camera composition and visual identity.",
  "camera_rule": "Do not use one identical camera position for every planet.",
  "information_rule": "Keep information concise and reveal it progressively.",
  "interaction_rule": "Allow the user to rotate or interact with the planet when appropriate.",
  "transition_rule": "Camera should physically travel from one planet to the next rather than instantly switching."
},
"planets": [
  {
    "name": "Mercury",
    "shot": "close-up",
    "screen_coverage": "35-45%",
    "headline": "THE SWIFT MESSENGER",
    "visual_focus": ["craters", "rocky surface", "strong sunlight"]
  },
  {
    "name": "Venus",
    "shot": "medium close-up",
    "headline": "A VEILED WORLD OF CRUSHING HEAT",
    "visual_focus": ["thick atmosphere", "clouds", "yellow-orange atmosphere"]
  },
  {
    "name": "Earth",
    "shot": "cinematic half-planet shot",
    "headline": "THE ONLY KNOWN WORLD WITH LIFE",
    "visual_focus": ["blue oceans", "clouds", "atmosphere", "Moon"],
    "composition": "Earth should be positioned slightly off-center with information on the opposite side."
  },
  {
    "name": "Mars",
    "shot": "angled close-up",
    "headline": "THE RUSTY RED FRONTIER",
    "visual_focus": ["red surface", "canyons", "mountains", "dust"]
  },
  {
    "name": "Jupiter",
    "shot": "medium-wide",
    "headline": "KING OF THE PLANETS",
    "visual_focus": ["Great Red Spot", "cloud bands", "large scale", "moons"]
  },
  {
    "name": "Saturn",
    "shot": "hero cinematic shot",
    "headline": "THE RINGED JEWEL",
    "visual_focus": ["rings", "planetary atmosphere", "dramatic lighting"],
    "composition": "Use a diagonal composition with the rings extending across the frame."
  },
  {
    "name": "Uranus",
    "shot": "medium close-up",
    "headline": "THE TILTED ICE GIANT",
    "visual_focus": ["cyan color", "axial tilt", "subtle rings"]
  },
  {
    "name": "Neptune",
    "shot": "distant cinematic close-up",
    "headline": "THE DISTANT WINDSWEPT BLUE",
    "visual_focus": ["deep blue atmosphere", "storm systems", "dark space"]
  }
],
"information_ui": {
  "style": "small cinematic glass panel",
  "position": "opposite the main 3D object whenever possible",
  "maximum_information": "3-5 key facts at one time",
  "animation": "fade + translate 20-30px",
  "exit_animation": "fade out + slight upward movement",
  "avoid": [
    "large opaque boxes",
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      "long paragraphs",
      "covering the planet",
      "too many statistics simultaneously"
    ]
  },
  "navigation": {
    "style": "minimal floating bottom navigation",
    "sections": [
      "Galaxy", "Milky Way", "Solar System", "Sun",
      "Mercury", "Venus", "Earth", "Mars",
      "Jupiter", "Saturn", "Uranus", "Neptune"
    ],
    "behavior": [
      "Highlight active section.",
      "Smooth scroll to selected section.",
      "Keep navigation unobtrusive.",
      "Allow keyboard navigation."
    ]
  },
  "scroll_behavior": {
    "primary_function": "Scroll controls the camera journey.",
    "easing": "smooth cinematic interpolation",
    "camera_motion": "continuous rather than abrupt",
    "rule": "Each section should have a clear start camera position and end camera position.",
    "transitions": "Blend camera position, camera target, object scale and lighting between sections."
  },
  "animation_guidelines": {
    "text_in": "opacity 0 to 1 + translateY 30px to 0",
    "text_out": "opacity 1 to 0 + translateY 0 to -20px",
    "3d_objects": "slow natural rotation",
    "particles": "subtle movement",
    "camera": "smooth interpolation",
    "interaction": "responsive but never overly fast",
    "reduced_motion": "respect prefers-reduced-motion and provide a manual reduced-motion toggle"
  },
  "responsive_design": {
    "desktop": "cinematic full-screen 3D experience",
    "tablet": "reduce particle count and simplify UI",
    "mobile": [
      "reduce 3D object scale",
      "move information below or beside the object depending on available space",
      "reduce particle density",
      "maintain readable typography",
      "maintain touch interaction"
    ]
  },
  "performance": {
    "three_js": true,
    "limit_pixel_ratio": 2,
    "reduce_particles_on_mobile": true,
    "lazy_load_heavy_assets": true,
    "avoid_unnecessary_dom_animation": true,
    "maintain_smooth_scroll": true
  },
  "accessibility": {
    "keyboard_navigation": true,
    "reduced_motion": true,
    "visible_focus_states": true,
    "aria_labels": true,
    "sufficient_text_contrast": true
  },
  "final_visual_direction": {
    "keywords": [
      "cinematic", "immersive", "premium", "minimal",
      "scientific", "futuristic", "deep space",
      "interactive", "editorial", "3D storytelling"
    ],
    "reference_feeling": "An interactive space documentary where the user physically travels from the Milky Way into the Solar System and
    "most_important_rule": "Never make the website feel like a static webpage placed on top of a 3D model. The 3D environment, camera mov
  },
  "output_requirements": {
    "deliverable": "Complete working HTML file",
    "include": [
      "HTML", "CSS", "JavaScript", "Three.js",
      "responsive design", "scroll-based camera system",
      "interactive planets", "animated information",
      "navigation", "reduced motion mode"
    ],
    "code_quality": [
      "well structured",
      "commented",
      "easy to modify",
      "avoid unnecessary dependencies"
    ]
  }
}
```